HARDWARE SPECIFICATION

Flight Controller — Prototype v1

Parameter	Value
Document ID	HW-SPEC-PROTO-001
Version	1.0 — Draft
Date	February 2026
Parent document	SYS-REQ-001
Status	Draft
Certification	Not certified — internal validation prototype only

1. Purpose

This document specifies the hardware requirements for the design and manufacture of the Flight Controller (FC) Prototype v1. It is intended for the hardware engineer responsible for PCB design, layout, and assembly.

The prototype serves to validate hardware choices prior to the certified DAL-B production design. It is not subject to DO-254 certification at this stage.

2. System Architecture

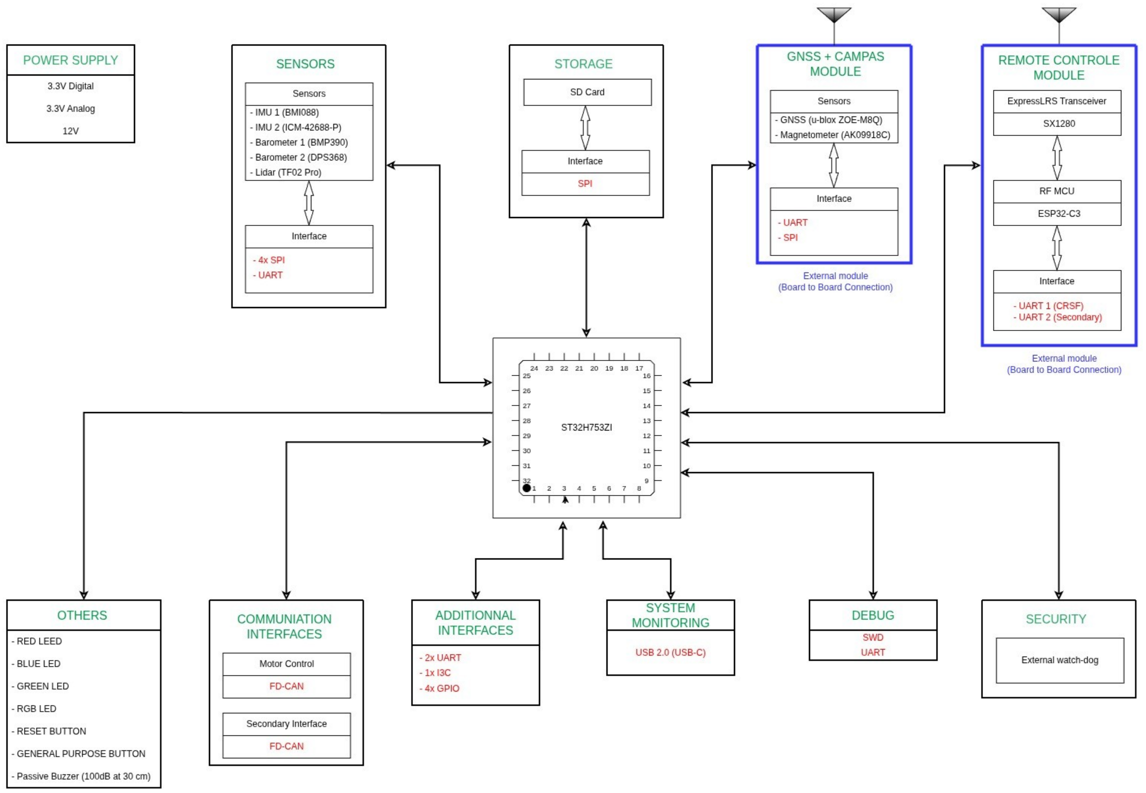
The FC system consists of three separate PCBs interconnected via board-to-board connectors:

Board	PCB Size	Role
FCU (Main Board)	100 × 100 mm	MCU, inertial sensors, power interfaces, communication
GNSS + Compass Module	External module	u-blox ZOE-M8Q + AK09918C
Remote Control Module	External module	ExpressLRS SX1280 + ESP32-C3

- GNSS and RC modules connect to the FCU via board-to-board stacking connectors (left and right sides of the FCU)
- All three PCBs are 4-layer

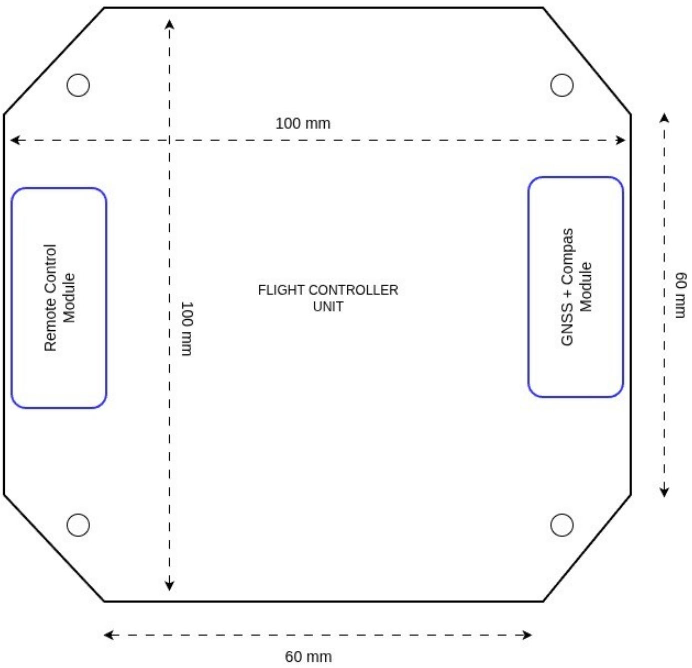
2.1 Block Diagram

Figure 1 — FC System Block Diagram



2.2 Mechanical Layout

Figure 2 — FCU Board Layout (100 × 100 mm octagonal, modules on left and right sides)



3. Fixed Components

The following critical components are imposed. No substitution without prior validation.

3.1 FCU — Main Board

Microcontroller

Parameter	Value
MCU	ST32H753ZI
Architecture	ARM Cortex-M7 @ 480 MHz — hardware double-precision FPU
Justification	16D EKF @ 1 kHz — SYS-REQ-F-010, SYS-REQ-F-013

Inertial Sensors — Redundancy Required (SYS-REQ-F-001)

Reference	Component	Bus	Rate	Notes
IMU 1	BMI088	SPI	≥ 1 kHz	Gyro + Accel
IMU 2	ICM-42688-P	SPI	≥ 1 kHz	Gyro + Accel
Barometer 1	BMP390	SPI	≥ 50 Hz	
Barometer 2	DPS368	SPI	≥ 50 Hz	

IMPORTANT: Both IMUs and both barometers must be mounted on vibration isolation pads (soft-mount).

Distance Sensor

Reference	Component	Bus	Rate
Lidar	TF02 Pro	UART	≥ 50 Hz — SYS-REQ-F-001

3.2 GNSS + Compass Module (external board)

Reference	Component	Interface to FCU	Rate
GNSS	u-blox ZOE-M8Q	UART	≥ 5 Hz — SYS-REQ-F-001
Magnetometer	AK09918C	SPI	≥ 50 Hz — SYS-REQ-F-001

- External GNSS antenna via U.FL or SMA connector
- Module positioned on the right/left side of the FCU (see Figure 2)

3.3 Remote Control Module (external board)

Reference	Component	Interface to FCU	Notes
RF Transceiver	SX1280	SPI (via ESP32-C3)	ExpressLRS — SYS-REQ-IF-001
RF MCU	ESP32-C3	UART CRSF to FCU	Primary CRSF protocol

- FCU interfaces: UART 1 (CRSF primary) + UART 2 (secondary)
- Module positioned on the left/right side of the FCU (see Figure 2)

4. Power Supply

The FCU has no direct access to the battery. All power rails are supplied by the propulsion board via dedicated JST-PH 2-pin connectors (one per rail). Maximum current per rail: 2A.

Rail	Voltage	Max Current	Connector	Usage
3.3V Digital	3.3V regulated	2A	JST-PH 2-pin	MCU, digital sensors, logic
3.3V Analog	3.3V regulated	2A	JST-PH 2-pin	Analog sensors, MCU ADC — filtered, isolated from digital rail
12V Regulated	12V regulated	2A	JST-PH 2-pin	Buzzer, peripherals

NOTE: The FCU does not include any DC-DC converters or LDOs for these rails. The three JST-PH connectors are the sole power inputs from the propulsion board. Current monitoring is not available on the FCU side.

Battery voltage and current monitoring (SYS-REQ-PWR-001/002) are provided by the propulsion board and transmitted to the FCU via CAN bus.

5. PCB Specifications

Parameter	Specification
FCU dimensions	100 × 100 mm — octagonal shape (chamfered corners)
Module dimensions	Not specified
Number of layers	4 layers — all boards
Mounting holes FCU	4× M3 — one at each corner
Module mounting	Board-to-board stacking connectors, lateral overhang

6. Required Interfaces

6.1 Sensor Interfaces (FCU)

Bus	Qty	Assignment	Notes
SPI	4×	IMU1, IMU2, Baro1, Baro2	Dedicated CS per sensor
UART	1×	Lidar TF02 Pro	Verify TF02 operating voltage (5V or 3.3V)

6.2 Motor Control Interface (FCU)

Interface	Qty	Usage
FD-CAN	2×	Motor control (CAN Bus) — SYS-REQ-F-020

NOTE: FD-CAN is selected for motor control. ESCs are CAN-compatible.

6.3 Additional Communication Interfaces (FCU)

Interface	Qty	Usage
UART	2×	Additional interfaces (e.g. telemetry, expansion)
I3C/I2C	1×	Additional interface
GPIO	4×	General purpose

6.4 System Interfaces (FCU)

Interface	Qty	Usage
USB-C	1×	System monitoring (USB 2.0) — SYS-REQ-NF-030
SWD	1×	MCU programming and debug
UART debug	1×	MCU serial debug
SD Card	1×	Flight data logging — SYS-REQ-IF-011 (SPI bus)

6.5 Board-to-Board Interfaces (FCU ↔ Modules)

Connection	Direction	Signals
FCU ↔ GNSS+Compass	Board-to-board	UART, SPI, 3.3V Digital, GND
FCU ↔ RC Module	Board-to-board	UART1 (CRSF), UART2, 3.3V Digital, GND

Connector reference, pitch, and stack height are left to the hardware engineer's judgment based on mechanical constraints and availability.

7. Visual and Audio Indicators

Component	Specification	SYS-REQ
Red LED	1× red LED — critical state	SYS-REQ-ARM-003
Blue LED	1× blue LED — info state	SYS-REQ-ARM-003
Green LED	1× green LED — armed / OK	SYS-REQ-ARM-003
RGB LED	1× RGB LED — general status	SYS-REQ-ARM-003
Passive Buzzer	100 dB at 30 cm	SYS-REQ-ARM-003

8. Physical Controls

Element	Specification
RESET button	1× push button — MCU reset
General purpose btn	1× push button — configurable (e.g. manual arming)

9. Hardware Security

Element	Specification
External watchdog	1× hardware watchdog — 500 ms timeout — SYS-REQ-NF-023
Justification	System reset if not refreshed by MCU — protects against software lockup

10. Timing Requirements

These constraints are critical for the flight control loop. The hardware engineer must ensure all interfaces support the following data rates.

Loop / Sensor	Rate / Period	Reference
IMU (gyro + accel)	≥ 1 kHz	SYS-REQ-F-013, SYS-REQ-NF-010
Fast control loop (rate)	1 ms — max jitter 0.2 ms	SYS-REQ-NF-010
Medium control (attitude)	5 ms — max jitter 0.5 ms	SYS-REQ-NF-010
Barometer	≥ 50 Hz	SYS-REQ-F-001
GNSS	≥ 5 Hz	SYS-REQ-F-001
Telemetry / Logging	≥ 1 Hz / ≥ 10 Hz	SYS-REQ-IF-005, SYS-REQ-IF-012

11. Deliverables

#	Deliverable	Details
1	Schematic	PDF + source file (KiCad)
2	PCB layout	Gerber files + drill files
3	BOM	Manufacturer references, quantities, suppliers
4	Design notes	Key decisions, known limitations, simulation results

12. Acceptance Criteria

The prototype is considered acceptable when all of the following checks pass:

#	Criterion	Reference
1	MCU boots successfully in < 5 seconds	SYS-REQ-SM-002
2	IMU 1 (BMI088) readable at 1 kHz via SPI	SYS-REQ-F-001
3	IMU 2 (ICM-42688-P) readable at 1 kHz via SPI	SYS-REQ-F-001
4	Barometer 1 (BMP390) and Barometer 2 (DPS368) readable at 50 Hz	SYS-REQ-F-001
5	GNSS (ZOE-M8Q) — fix acquired, data @ 5 Hz	SYS-REQ-F-001
6	Magnetometer (AK09918C) readable at 50 Hz	SYS-REQ-F-001
7	Lidar (TF02 Pro) readable via UART at 50 Hz	SYS-REQ-F-001
8	SD Card — read/write functional	SYS-REQ-IF-011
9	FD-CAN — frame transmitted and received on both buses	SYS-REQ-F-020
10	USB-C — serial communication active	SYS-REQ-NF-030
11	External watchdog — system reset triggered if not refreshed in 500ms	SYS-REQ-NF-023
12	All three power rails stable under load (3.3V dig, 3.3V ana, 12V)	SYS-REQ-PWR-001
13	RGB LED + red/blue/green LEDs — all functional	SYS-REQ-ARM-003
14	Buzzer — tone audible at 30 cm	SYS-REQ-ARM-003
15	IMU loop jitter < 0.2 ms (measured via oscilloscope or GPIO toggle)	SYS-REQ-NF-010

13. Design Freedom

The following design decisions are left to the hardware engineer's judgment:

- Board-to-board connector selection (pitch, stack height, manufacturer reference)
- External connectors type and reference (JST-GH recommended but not mandatory)
- Component placement and PCB routing
- Analog power rail filtering strategy (EMI filters, ferrite beads, decoupling)
- Passive component footprints

Revision History

Version	Date	Author	Changes
1.0	Feb. 2026	BASSINGA Brisel Andresse	Initial release